

Introducing ψ -Ordered Complexity Classes for Sequential Algorithms

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Abstract

This paper introduces a novel asymptotic notation, ψ , designed to capture the *ordering constraint* between two functions or computational layers. While classical notations such as O , Ω , and Θ describe the growth rate of a *single* function, the proposed ψ -notation provides a formal guarantee that one layer (F_1) is strictly bounded above by a subsequent layer (F_2) within the same asymptotic class.

1 General Definition

Let $F_1, F_2 : \mathbb{N} \rightarrow \mathbb{R}^+$ be a pair of non-negative functions. We define the ordered pair (F_1, F_2) to belong to the complexity class $\psi(g(n))$ if there exist positive real constants c_1, c_2, c_3 and a threshold $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$, the following chain of inequalities holds:

$$c_1 \cdot g(n) \leq F_1(n) \leq c_2 \cdot g(n) \leq F_2(n) \leq c_3 \cdot g(n) \quad (1)$$

This definition inherently guarantees that $F_1(n) \leq F_2(n)$ asymptotically, making it ideal for pipeline architectures where upstream processes must never overload downstream processes.

2 Special Cases of Complexity Classes

By substituting $g(n)$ with common growth functions, the ψ -notation provides fine-grained ordered complexity classes:

2.1 $\psi(1)$: Constant Ordered Time

For $g(n) = 1$, the condition reduces to:

$$c_1 \leq F_1(n) \leq c_2 \leq F_2(n) \leq c_3 \quad (2)$$

This represents the optimal efficiency class, where both layers execute in $O(1)$ time with guaranteed ordered resource allocation.

2.2 $\psi(\log N)$: Logarithmic Ordered Complexity

For $g(n) = \log N$:

$$c_1 \cdot \log N \leq F_1(N) \leq c_2 \cdot \log N \leq F_2(N) \leq c_3 \cdot \log N \quad (3)$$

This characterizes ordered algorithms with divide-and-conquer behavior, ensuring the first stage scales strictly slower than or equal to the second.

2.3 $\psi(N)$: Linear Ordered Complexity

For $g(n) = N$:

$$c_1 \cdot N \leq F_1(N) \leq c_2 \cdot N \leq F_2(N) \leq c_3 \cdot N \quad (4)$$

This models linear systems where the input size directly scales the workload, and the downstream stage is provisioned to always handle at least the peak load of the upstream stage.

2.4 $\psi(N \log N)$: Log-Linear Ordered Complexity

For $g(n) = N \log N$:

$$c_1 \cdot N \log N \leq F_1(N) \leq c_2 \cdot N \log N \leq F_2(N) \leq c_3 \cdot N \log N \quad (5)$$

This class is particularly relevant for high-performance sorting algorithms and divide-and-conquer merging operations where strict phase ordering is critical.

2.5 $\psi(N^2)$: Quadratic Ordered Complexity

For $g(n) = N^2$:

$$c_1 \cdot N^2 \leq F_1(N) \leq c_2 \cdot N^2 \leq F_2(N) \leq c_3 \cdot N^2 \quad (6)$$

This applies to nested iterative algorithms, guaranteeing that the inner processing layer (e.g., heavy computation) remains asymptotically dominant while the outer layer (e.g., data retrieval) remains strictly bounded below it.

2.6 Generalization to Arbitrary Functions

The notation is not restricted to these forms. For any well-defined positive function $h(n)$, one can naturally extend the definition to $\psi(h(n))$ as long as the chain in (1) holds with $g(n) = h(n)$. This flexibility allows the ψ -notation to adapt to any algorithmic complexity curve.

3 Conclusion

The ψ -notation complements the standard Landau notations by introducing a *structural ordering* dimension to asymptotic analysis. It serves as a powerful tool for system architects and algorithm designers who require formal mathematical guarantees on the balance between sequential computational stages.